Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

Cloud Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, building multi-account AWS platforms for enterprise financial-services and technology customers. Currently building the Terraform foundation for RWS's data-centre exit and a second landing zone: Organizations OUs with Control Tower baselines, SCPs, tag policies, and two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect. Worked on CCMv4.0-aligned landing zone controls at State Street (1,000+ preventative controls, team result) and modernized cloud-native IaC at DTCC. Terraform module and Landing Zone specialist; 11 upstream PRs merged (axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others). 5x AWS certified including Solutions Architect – Associate.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (~90%) by building the AWS foundation for a data-centre exit and a second landing zone entirely in Terraform.
  - Delivered two Transit Gateway hubs in separate AWS Regions with Network Firewall inspection and VPN/Direct Connect; self-service onboarding gives each new workload its route table within 15 minutes and firewall access by tag in about a minute.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines with a security baseline (IMDSv2, EBS encryption, no default VPC).
  - Won approval for AWS IPAM by measuring real billing in a sandbox (prefix delegation, about 94% lower projected cost).
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement with delegated admin in the Audit account via an AFT customization. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Provisioned end-to-end SageMaker MLOps infrastructure via Terraform + CodePipeline – blue-green endpoints with auto-scaling, CloudFront frontend, API Gateway + Lambda inference, drift-triggered retraining, and alarm-driven automatic rollback. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).
- **Internal Impact:** Published two more AWS samples in Terraform, on Organizations governance and a Bedrock AgentCore agent (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. MLOps SME Program Graduate. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). 2 org-wide tech talks on Terraform testing and GitOps gating. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases with X-Ray and CloudWatch distributed tracing.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS for continuous code quality analysis.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – Terraform for a Bedrock AgentCore agent (71 resources in 12 files): MCP tools behind AgentCore Gateway with a default-deny Cedar policy, PROD promotion only after an evaluation passes, CloudWatch alarms, a canary and a budget. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

Other Notable Projects: LeetCode Rating Predictor (train/serve skew fix, NumPy serving), itr-agent (deterministic tax engine, 151 tests), farm-stack-aws-ecs (Docker + GitHub Actions OIDC deploy to ECR/ECS), GitScope (Chrome Web Store extension), Instagram Likes Leaderboard (browser-console tool).

Open Source Contributions (11 merged upstream)

- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

AWS Multi-Account & Governance: Organizations, OU Design, SCPs, Tag Policies, Landing Zone, Control Tower, AWS Config, Security Hub, IAM Identity Center, Service Catalog, CloudTrail, GuardDuty
Networking & Hybrid Connectivity: Transit Gateway (hub-and-spoke), VPC, VPN, Direct Connect, PrivateLink, Route 53 Resolver, Centralized Egress Inspection, Network Firewall
Compute & Application Services: ECS Fargate, Lambda, API Gateway, CloudFront, ALB/NLB, EventBridge, Step Functions, SageMaker
Infrastructure as Code: Terraform (modules, tftest, PyTest), CloudFormation, HCL, Multi-Account Strategy, Docker
CI/CD & Automation: GitHub Actions, CodePipeline, CodeBuild, CodeDeploy, Blue-Green Deployments, GitOps, Policy-as-Code
Security & Compliance: CCMv4.0, Preventative SCPs, KMS, Least-Privilege IAM, Config Rules, Security Hub Standards, Amazon Q
Languages: HCL, Python, TypeScript, JavaScript, Bash, SQL